

TOP 100 INTERVIEW QUESTIONS

Campus Placement • CSE / Software Engineering

RESOURCE	DETAIL
Questions	100 interview questions + detailed answers
Coverage	HR • Behavioral • OOP • DSA • DBMS • OS • Networks • Web • Coding
Audience	BTech / CSE students and freshers
Style	Colorful, structured, placement-focused
Answer approach	Interview-ready explanations + practical tips

Interview rule: Do not memorize every answer word-for-word. Understand the idea, then answer naturally using your own project and experience.

Best preparation method: Practice answers aloud, keep technical explanations concise, and be ready for follow-up questions on every technology listed on your resume.

INTERVIEW ANSWER FRAMEWORK

Question type	Recommended structure
Tell me about yourself	Present → skills → projects → achievement → role fit
Technical concept	Definition → example → complexity/trade-off → use case
Project question	Problem → architecture → your role → challenge → result
Behavioral question	STAR: Situation → Task → Action → Result
Coding problem	Clarify → brute force → optimize → complexity → code → test
Unknown question	State what you know → reason carefully → explain how you would verify

Important: If you mention Java, React, Angular, Node.js, MySQL, AWS, Git, etc. on your resume, prepare basic interview questions for each one. Interviewers often drill into whatever you claim to know.

HR & PERSONAL

Frequently Asked HR Questions

Q1. Tell me about yourself.

Answer: Use a 60–120 second structure: present education/role → strongest technical skills → 1–2 relevant projects → achievement/learning → why you are interested in this role. Avoid unrelated personal history.

Interview Tip: Keep it customized to the job description rather than reciting your resume.

Q2. Why should we hire you?

Answer: Connect three things: the role's needs, your evidence, and your attitude. Example: "I have a solid foundation in Java, web development and SQL, I have applied them in projects, and I learn quickly when a new technology is required."

Interview Tip: Give evidence, not generic claims such as "I am hardworking."

Q3. Why do you want to join our company?

Answer: Mention something specific about the company, role, product or engineering culture, then connect it to your career goals and what you can contribute.

Interview Tip: Research the company before the interview; avoid saying only "because it is a reputed company."

Q4. What are your strengths?

Answer: Choose 2–3 job-relevant strengths and support each with a short example. For a software role, examples can include problem solving, consistency, learning ability, communication or ownership.

Interview Tip: Strengths are more credible when backed by a project, result or situation.

Q5. What is your weakness?

Answer: Choose a genuine but manageable weakness, explain what you are doing to improve it, and show measurable progress. Example: "I used to spend too long perfecting small UI details, so I now time-box them and prioritize functionality first."

Interview Tip: Do not use "I have no weaknesses" or a fake strength disguised as a weakness.

Q6. Where do you see yourself in five years?

Answer: Show realistic growth: becoming a strong software engineer, taking ownership of larger modules, improving system-design and technical skills, and contributing to team outcomes.

Interview Tip: Focus on capability and responsibility, not only job titles.

Q7. Why did you choose Computer Science?

Answer: Explain your interest in technology and problem solving, then mention how coursework and projects confirmed that interest.

Interview Tip: Make the answer personal and connected to your actual work.

Q8. What motivates you?

Answer: A strong answer can combine learning, solving difficult problems, seeing measurable results, and contributing to a team.

Interview Tip: Avoid making salary your only motivation.

Q9. Are you willing to relocate?

Answer: Answer honestly. If you are flexible, say so clearly; if you have constraints, explain them professionally and mention what locations or conditions work for you.

Interview Tip: Never promise something you cannot actually accept.

Q10. What are your salary expectations?

Answer: For a fresher, a safe approach is to say you are open to the company's standard package for the role and are primarily focused on the opportunity, learning and responsibilities.

Interview Tip: If the company asks for a number, give a researched and reasonable range rather than an arbitrary figure.

BEHAVIORAL

STAR Method & Workplace Situations

Q11. Tell me about a time you faced a difficult challenge.

Answer: Use STAR: Situation, Task, Action, Result. Clearly explain what made the situation difficult and focus most of the answer on what you personally did.

Q12. Describe a project you are proud of.

Answer: Explain the problem, your role, technology used, one difficult technical decision, and the result. Be ready for follow-up questions about every technology you mention.

Q13. Tell me about a time you made a mistake.

Answer: Choose a real but non-catastrophic mistake, take responsibility, explain the corrective action and state what process you changed to prevent recurrence.

Q14. How do you handle pressure?

Answer: Prioritize tasks, break work into smaller actions, communicate early about blockers, and protect accuracy on critical work.

Q15. How do you handle conflict with a teammate?

Answer: Listen first, clarify the technical or project issue, use evidence, discuss alternatives respectfully, and escalate only when necessary.

Q16. Describe a time you showed leadership.

Answer: Leadership does not require a formal title. Explain how you coordinated people, took ownership, removed a blocker, or helped the team reach a result.

Q17. How do you handle criticism?

Answer: Listen without becoming defensive, identify the useful part of the feedback, apply it, and follow up when appropriate.

Q18. Tell me about a time you learned something quickly.

Answer: Describe the technology or concept, the time constraint, your learning method, how you practiced it, and where you applied it.

Q19. What do you do when you do not know an answer?

Answer: Be honest, state what you know, reason from fundamentals if possible, and explain how you would verify the missing information.

Q20. How do you prioritize multiple tasks?

Answer: Rank tasks by urgency, impact, dependencies and deadlines. Communicate trade-offs when everything cannot be completed simultaneously.

TECHNICAL

Object-Oriented Programming & Java

Q21. What is OOP?

Answer: Object-Oriented Programming organizes software around objects containing state and behavior. The common pillars are encapsulation, abstraction, inheritance and polymorphism.

Q22. What is encapsulation?

Answer: Encapsulation bundles data and methods and controls access to the object's internal state, commonly through access modifiers and public methods.

Q23. What is abstraction?

Answer: Abstraction exposes essential behavior while hiding unnecessary implementation details. Interfaces and abstract classes are common mechanisms.

Q24. What is inheritance?

Answer: Inheritance allows a class to acquire or extend properties and behavior from another class. It promotes reuse but should be used when there is a genuine is-a relationship.

Q25. What is polymorphism?

Answer: Polymorphism allows the same interface or method call to behave differently for different object types. Overloading is compile-time polymorphism; overriding is runtime polymorphism in languages such as Java.

Q26. Overloading vs overriding?

Answer: Overloading uses the same method name with different parameter lists in the same class. Overriding occurs when a subclass provides a new implementation of an inherited method with the same signature.

Q27. Interface vs abstract class in Java?

Answer: An interface defines a contract and supports multiple interface inheritance. An abstract class can contain state, constructors and both abstract and concrete methods. The choice depends on whether you need a contract or shared base implementation.

Q28. What is a constructor?

Answer: A constructor initializes an object when it is created. It has the class name in Java and no return type.

Q29. What is the difference between == and equals() in Java?

Answer: For objects, == compares references, while equals() is intended to compare logical equality when the class overrides it appropriately.

Q30. Why is String immutable in Java?

Answer: Immutability improves safety and allows strings to be safely shared, cached and used in hash-based collections. Operations that appear to modify a String create a new object.

Data Structures & Algorithms

Q31. Array vs Linked List?

Answer: Arrays provide contiguous indexed storage and $O(1)$ random access. Linked lists use nodes connected by links; insertion/deletion can be efficient when a node position is known, but random access is $O(n)$.

Q32. What is a stack?

Answer: A stack follows LIFO: Last In, First Out. Common operations are push, pop and peek. It is used in recursion, expression evaluation and undo operations.

Q33. What is a queue?

Answer: A queue follows FIFO: First In, First Out. Common operations are enqueue and dequeue. It is used in scheduling, buffering and breadth-first search.

Q34. What is a hash table?

Answer: A hash table maps keys to values using a hash function. Average lookup, insertion and deletion are $O(1)$, assuming a good distribution and controlled collisions.

Q35. What is binary search?

Answer: Binary search repeatedly halves the search interval in a sorted collection. Its time complexity is $O(\log n)$.

Q36. What is the difference between BFS and DFS?

Answer: BFS explores level by level, typically using a queue; DFS explores deeply, typically using a stack or recursion. BFS is useful for shortest paths in unweighted graphs.

Q37. What is a binary search tree?

Answer: A BST is a binary tree where keys in the left subtree are smaller and keys in the right subtree are larger, subject to the chosen duplicate policy. Average search can be $O(\log n)$, but a skewed BST can become $O(n)$.

Q38. What is recursion?

Answer: Recursion is when a function solves a problem by calling itself on a smaller instance. It requires a correct base case and can use call-stack memory.

Q39. What is time complexity?

Answer: Time complexity describes how an algorithm's running time grows with input size. Big-O notation expresses an upper-bound growth rate such as $O(n)$, $O(\log n)$ or $O(n^2)$.

Q40. What is space complexity?

Answer: Space complexity measures additional memory required as input size grows, including auxiliary data structures and sometimes recursion stack space depending on the convention.

DBMS & SQL

Q41. What is a primary key?

Answer: A primary key uniquely identifies each row in a table. It must be unique and cannot contain NULL values.

Q42. What is a foreign key?

Answer: A foreign key is a column or set of columns that references a key in another table, helping enforce referential integrity.

Q43. What is normalization?

Answer: Normalization organizes relational data to reduce redundancy and update anomalies. Common normal forms include 1NF, 2NF and 3NF.

Q44. What is a JOIN?

Answer: A JOIN combines rows from tables using a related condition. Common types are INNER JOIN, LEFT JOIN, RIGHT JOIN and FULL OUTER JOIN.

Q45. INNER JOIN vs LEFT JOIN?

Answer: INNER JOIN returns only matching rows from both sides. LEFT JOIN returns every row from the left table and matching rows from the right; unmatched right-side values become NULL.

Q46. WHERE vs HAVING?

Answer: WHERE filters rows before grouping, while HAVING filters groups after GROUP BY and is commonly used with aggregate conditions.

Q47. What is an index?

Answer: An index is an auxiliary data structure that speeds up data retrieval for selected queries, at the cost of extra storage and additional work on inserts/updates/deletes.

Q48. DELETE vs TRUNCATE vs DROP?

Answer: DELETE removes selected rows and can use a WHERE clause. TRUNCATE removes all rows from a table with less row-by-row logging in many systems. DROP removes the table definition and its data.

Q49. What is ACID?

Answer: ACID describes transaction properties: Atomicity, Consistency, Isolation and Durability.

Q50. Write SQL to find the second-highest salary.

Answer: One common approach is: `SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);` This returns the second distinct-highest salary.

Operating Systems

Q51. Process vs thread?

Answer: A process is an independent execution environment with its own address space. Threads are execution units within a process and typically share the process's memory.

Q52. What is a context switch?

Answer: The OS saves the state of one running task and loads the state of another so the CPU can switch execution.

Q53. What is deadlock?

Answer: Deadlock is a state where processes wait indefinitely for resources held by one another. The classic necessary conditions are mutual exclusion, hold and wait, no preemption and circular wait.

Q54. What is virtual memory?

Answer: Virtual memory gives processes an abstraction of a large address space and can use disk-backed mechanisms such as paging when physical memory is insufficient.

Q55. What is paging?

Answer: Paging divides virtual memory into fixed-size pages and physical memory into frames. Page tables map virtual pages to physical frames.

Q56. What is CPU scheduling?

Answer: CPU scheduling selects which ready process/thread should run. Algorithms include FCFS, SJF, Round Robin and priority scheduling.

Q57. What is a race condition?

Answer: A race condition occurs when the result depends on the timing or interleaving of concurrent operations on shared state.

Q58. What is a semaphore?

Answer: A semaphore is a synchronization primitive used to control access to shared resources or coordinate concurrent tasks.

Computer Networks & Web

Q59. What is the OSI model?

Answer: The OSI model has seven layers: Physical, Data Link, Network, Transport, Session, Presentation and Application.

Q60. TCP vs UDP?

Answer: TCP is connection-oriented and provides reliable, ordered delivery with congestion and flow control. UDP is connectionless with lower overhead but does not guarantee delivery or order.

Q61. What is an IP address?

Answer: An IP address identifies a network interface or endpoint at the network layer. IPv4 uses 32 bits; IPv6 uses 128 bits.

Q62. What is DNS?

Answer: DNS translates human-readable domain names into IP addresses and other DNS records.

Q63. HTTP vs HTTPS?

Answer: HTTPS is HTTP carried over TLS, providing encryption, server authentication and integrity protection for the connection.

Q64. What is a port number?

Answer: A port identifies a logical service endpoint associated with network traffic. TCP and UDP use port numbers to direct data to applications.

Q65. What is the difference between GET and POST?

Answer: GET is primarily used to retrieve a resource and should be safe/idempotent in typical HTTP semantics. POST is commonly used to submit data or request creation/processing and is not generally idempotent.

Q66. What is an API?

Answer: An API is a defined interface through which software components communicate. A web API commonly exposes resources or operations over HTTP.

Projects, Web Development & APIs

Q67. Explain your project in one minute.

Answer: Use: problem → users → key features → architecture/stack → your contribution → result. Keep it understandable before discussing implementation details.

Q68. Why did you choose your technology stack?

Answer: Connect each technology to a requirement. For example, React for component-based UI, Node/Express for APIs, and MySQL for relational data. Be ready to justify alternatives.

Q69. How does a frontend communicate with a backend?

Answer: The frontend sends HTTP requests to backend API endpoints. The backend validates input, applies business logic, accesses data/services, and returns a response such as JSON.

Q70. What is REST?

Answer: REST is an architectural style for networked applications that commonly uses resources, standard HTTP methods, stateless requests and representations such as JSON.

Q71. What is JWT?

Answer: JSON Web Token is a compact token format often used to represent claims for authentication/authorization. A server can verify a signed token and use its claims to identify the requester.

Q72. Authentication vs authorization?

Answer: Authentication answers "Who are you?" Authorization answers "What are you allowed to do?" Login establishes identity; roles/permissions determine access.

Q73. What is CORS?

Answer: Cross-Origin Resource Sharing is a browser security mechanism controlling whether a web page from one origin can access resources from another origin. Servers can send headers to allow specific origins/methods.

Q74. How would you secure a login system?

Answer: Hash passwords with a strong password-hashing algorithm, use HTTPS, validate inputs, protect sessions/tokens, implement authorization, rate-limit sensitive endpoints and avoid exposing secrets.

Q75. How do you debug a project?

Answer: Reproduce the issue, inspect logs/errors, isolate the failing layer, form a hypothesis, test the smallest change, verify the fix, and add a regression test when practical.

Q76. What would you improve in your project if given more time?

Answer: Choose improvements with real engineering value: testing, performance, accessibility, security, observability, error handling, scalability or UX. Explain why the improvement matters.

PROBLEM SOLVING

Common Coding Interview Problems

Q77. How would you solve Two Sum?

Answer: Use a hash map storing value $\rightarrow$ index while scanning once. For each x , check whether $\text{target}-x$ is already present. Average time $O(n)$, space $O(n)$.

Q78. How do you reverse a linked list?

Answer: Iterate through nodes with prev, current and next pointers; reverse current.next to prev, then advance. Time $O(n)$, auxiliary space $O(1)$.

Q79. How do you find the maximum element in an array?

Answer: Scan once while maintaining a maximum variable. Time $O(n)$ and auxiliary space $O(1)$.

Q80. How do you check whether a string is a palindrome?

Answer: Use two pointers from both ends and compare characters while moving inward. Time $O(n)$, with $O(1)$ extra space for a simple implementation.

Q81. How do you detect a cycle in a linked list?

Answer: Floyd's tortoise-and-hare algorithm uses a slow pointer moving one step and a fast pointer moving two steps. If they meet, a cycle exists. Time $O(n)$, space $O(1)$.

Q82. How do you find the kth largest element?

Answer: Options include sorting in $O(n \log n)$, a min-heap of size k in $O(n \log k)$, or Quickselect with average $O(n)$. Choose based on constraints.

Q83. What is dynamic programming?

Answer: DP solves problems with overlapping subproblems and optimal substructure by storing results of smaller states, either top-down with memoization or bottom-up with tabulation.

Q84. Greedy vs dynamic programming?

Answer: Greedy makes the best local choice at each step and is correct only when the problem has the required greedy-choice property. DP systematically evaluates states and transitions.

Q85. How do you optimize an $O(n^2)$ algorithm?

Answer: First identify the bottleneck and constraints. Common improvements include hashing, sorting plus two pointers, binary search, prefix sums, heaps or a better algorithmic formulation.

Q86. What should you do before coding a problem?

Answer: Clarify inputs/outputs and constraints, test examples, consider edge cases, describe a brute-force approach, then derive and explain the optimized approach before coding.

TECHNICAL

Additional Software Engineering Questions

Q87. What is the difference between compile-time and runtime errors?

Answer: Compile-time errors are detected while code is being compiled, such as syntax or type errors. Runtime errors occur while the program executes, such as division by zero or a null reference.

Q88. What is exception handling in Java?

Answer: Exception handling manages abnormal conditions using mechanisms such as try, catch, finally, throw and throws so a program can handle failures without abruptly terminating.

Q89. ArrayList vs LinkedList in Java?

Answer: ArrayList provides fast indexed access and is generally preferred for read-heavy lists. LinkedList has node-based storage and can be useful for certain insertion/deletion patterns, but it has slower random access and higher memory overhead.

Q90. What is a static keyword in Java?

Answer: static makes a member belong to the class rather than to individual objects. Static fields are shared across instances, and static methods can be called without creating an object.

Q91. What is garbage collection?

Answer: Garbage collection automatically identifies objects that are no longer reachable and reclaims their memory. Java uses automatic garbage collection rather than requiring manual free/delete calls.

Q92. What is Git and why is it used?

Answer: Git is a distributed version-control system used to track code changes, collaborate through branches and merges, and maintain a history of project versions.

Q93. git merge vs git rebase?

Answer: Merge combines histories and creates a merge commit when needed. Rebase moves a branch's commits onto a new base and creates a cleaner linear history, but rewriting shared history should be avoided.

Q94. What is cloud computing?

Answer: Cloud computing provides on-demand computing resources such as servers, storage, databases and networking over a network, typically with elastic scaling and usage-based pricing.

Q95. What are IaaS, PaaS and SaaS?

Answer: IaaS provides infrastructure such as virtual machines and storage; PaaS provides a managed platform for building/deploying applications; SaaS provides complete software to end users.

Q96. What is an HTTP status code?

Answer: It is a numeric response code indicating the result of an HTTP request. Examples include 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found and 500 Internal Server Error.

Q97. What is JSON?

Answer: JSON is a lightweight text format commonly used for exchanging structured data between clients and servers. It represents objects, arrays, strings, numbers, booleans and null.

Q98. What is responsive web design?

Answer: Responsive design makes a web interface adapt to different screen sizes and devices using flexible layouts, media queries and appropriately sized content.

Q99. What is unit testing?

Answer: Unit testing verifies small isolated pieces of code, such as functions or classes, so defects can be detected early and changes can be made more safely.

Q100. What is the difference between authentication using sessions and tokens?

Answer: Session authentication stores server-side session state and sends a session identifier to the client. Token-based authentication sends a signed token containing claims that the server can validate; the right choice depends on architecture and security requirements.

FINAL INTERVIEW CHECKLIST

- ✓ Prepare a 60–120 second self-introduction.
- ✓ Know every project and technology listed on your resume.
- ✓ Revise OOP, DSA, DBMS/SQL, OS and Computer Networks.
- ✓ Practice at least 10 coding problems without looking at solutions.
- ✓ Prepare 4–5 STAR stories for challenges, teamwork, leadership and failure.
- ✓ Research the company, role, products and recent work before the interview.
- ✓ Prepare 2–3 intelligent questions to ask the interviewer.
- ✓ Carry your resume and required documents.
- ✓ Practice speaking clearly and explaining your thought process.
- ✓ Do not bluff technical knowledge; explain what you know and how you would learn what you do not.

Last-minute strategy: Revise your resume first, then OOP + SQL + DSA basics, then your projects, and finally HR/behavioral answers. For coding rounds, prioritize arrays, strings, hashing, linked lists, stacks/queues, binary search and basic recursion.